THE UK UNIVERSITY INTEGRATION BEE 2025/2026

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ROUND ONE SOLUTIONS

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1

$$\int_0^{\frac{\pi}{4}} \frac{1}{x^2+1}\,\mathrm{d}x$$

Answer: $\tan^{-1}\left(\frac{\pi}{4}\right)$.

Solution: Direct integration gives

$$\int_0^{\frac{\pi}{4}} \frac{1}{x^2+1}\,\mathrm{d}x = \tan^{-1}\left(\frac{\pi}{4}\right).$$

Be careful not to accidentally "simplify" that final result! □

2

$$\int_0^\infty \sin(\pi \lfloor x \rfloor)\, \mathrm{d}x$$

Answer: 0.

Solution: Splitting up the integral, we have that

$$\int_0^\infty \sin(\pi \lfloor x \rfloor)\, \mathrm{d}x = \sum_{k=0}^{\infty} \int_k^{k+1} \sin(\pi k)\, \mathrm{d}x = 0,$$

since every sine term is 0. □

3

$$\int_0^\infty \operatorname{sech}(2\ln(x))\,\mathrm{d}x$$

Answer: $\dfrac{\pi}{\sqrt{2}}$.

Solution: Write the hyperbolic cosecant in exponential form, and we have

$$\int_0^\infty \operatorname{sech}(2\ln(x))\,\mathrm{d}x = \int_0^\infty \frac{2}{x^2+x^{-2}}\,\mathrm{d}x = \int_0^\infty \frac{2}{\left(x-\frac{1}{x}\right)^2+2}\,\mathrm{d}x\,.$$

Using Glasser's master theorem (a.k.a. the $x-\dfrac{1}{x}\mapsto x$ substitution), this becomes

$$\int_0^\infty \frac{2}{x^2+2}\,\mathrm{d}x = \frac{\pi}{\sqrt{2}}.$$

□

4

$$\int_0^a e^{\frac{i\pi}{2}} \, \mathrm{d}z$$

Answer: ai.

Solution: Notice that $e^{\frac{i\pi}{2}} = i$ so we get

$$\int_0^a e^{\frac{i\pi}{2}} \, \mathrm{d}z = ai.$$

We just wanted an answer that was AI. □

5

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(x)}{1+\sqrt[x]{e}}\,\mathrm{d}x$$

Answer: 1.

Solution: Apply the border flip $x \mapsto -x$. We have

$$I := \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(x)}{1+\sqrt[x]{e}}\,\mathrm{d}x = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(x)}{1+\sqrt[-x]{e}}\,\mathrm{d}x\,.$$

Then we have, after combining fractions and simplifying,

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x)\,\mathrm{d}x = 2.$$

So $I = 1$. □

6

$$\int_0^1 \sqrt{-\ln(x)}\,\mathrm{d}x$$

Answer: $\frac{\sqrt{\pi}}{2}$.

Solution: Use the substitution $u = -\ln(x)$ to get

$$\int_0^1 \sqrt{-\ln(x)}\,\mathrm{d}x = \int_0^\infty \sqrt{u}e^{-u}\,\mathrm{d}u = \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

□

7

$$\int_0^1 \left(\{x\} + \{x\{x\}\} + \{x\{x\{x\}\}\} + \cdots - \frac{1}{x} \right) \mathrm{d}x$$

Answer: Does not converge (but has Cauchy principal value -1).

Solution: Since the integration bounds are 0 to 1, we may treat $\{x\} = x$ and consequently

$$\begin{aligned} &\int_0^1 \left(\{x\} + \{x\{x\}\} + \{x\{x\{x\}\}\} + \cdots - \frac{1}{x} \right) \mathrm{d}x \\ &= \int_0^1 \left(x + x^2 + x^3 + \cdots - \frac{1}{x} \right) \mathrm{d}x \\ &= \int_0^1 \left(\frac{1}{1-x} - 1 - \frac{1}{x} \right) \mathrm{d}x \,. \end{aligned}$$

Observe that the integral $\displaystyle\int_0^1 \left(\frac{1}{1-x} - \frac{1}{x} \right) \mathrm{d}x$ diverges.

Supposing that we instead take the Cauchy principal value of the integral, by King's rule we have

$$\int_\varepsilon^{1-\varepsilon} \left(\frac{1}{1-x} - \frac{1}{x} \right) \mathrm{d}x = 0,$$

for $\varepsilon > 0$ small. It follows that

$$\mathrm{P.V.} \int_0^1 \left(\frac{1}{1-x} - \frac{1}{x} \right) \mathrm{d}x = 0.$$

So the remaining integral is

$$\int_0^1 (-1) \, \mathrm{d}x = -1.$$

□

8

$$\int_0^\infty \left(1-\frac{x}{1}\left(1-\frac{x}{2}\left(1-\frac{x}{3}\left(1-\frac{x}{4}\left(\dots\right)\right)\right)\right)\right)\mathrm{d}x$$

Answer: 1.

Solution: Direct expansion of the nested products gives

$$\begin{aligned}
&\int_0^\infty \left(1-\frac{x}{1}\left(1-\frac{x}{2}\left(1-\frac{x}{3}\left(1-\frac{x}{4}\left(\dots\right)\right)\right)\right)\right)\mathrm{d}x \\
&\quad= \int_0^\infty \left(1-\frac{x}{1}+\frac{x^2}{2!}-\frac{x^3}{3!}+\dots\right)\mathrm{d}x \\
&\quad= \int_0^\infty e^{-x}\,\mathrm{d}x \\
&\quad= 1.
\end{aligned}$$

□

9

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{7x^7 - 5x^5 + 3x^3 - x + 1}{\cos^2(x)} \, \mathrm{d}x$$

Answer: 2.

Solution: This looks much scarier than it actually is. Observe that if n is odd, the function $\dfrac{x^n}{\cos^2(x)}$ is odd, and therefore the integral is simply

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{7x^7 - 5x^5 + 3x^3 - x + 1}{\cos^2(x)} \, \mathrm{d}x = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2(x) \, \mathrm{d}x = 2.$$

□

10

$$\int_0^{\frac{\pi}{2}} \frac{\tan^{-1}(b\sin(x))}{\sin(x)}\,\mathrm{d}x$$

Answer: $\dfrac{\pi}{2}\sinh^{-1}(b)$.

Solution: Traditionally, Problem 10 is a DUTIS problem. Let $I(b) = \displaystyle\int_0^{\frac{\pi}{2}} \frac{\tan^{-1}(b\sin(x))}{\sin(x)}\,\mathrm{d}x$. We have

$$\begin{aligned} I'(b) &= \int_0^{\frac{\pi}{2}} \frac{1}{b^2\sin^2(x)+1}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} \frac{\sec^2(x)}{b^2\tan^2(x)+\sec^2(x)}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} \frac{\sec^2(x)}{(b^2+1)\tan^2(x)+1}\,\mathrm{d}x\,. \end{aligned}$$

Let $u = \tan(x)$. Then the integral is

$$I'(b) = \int_0^{\infty} \frac{1}{(b^2+1)u^2+1}\,\mathrm{d}u = \frac{\pi}{2\sqrt{b^2+1}}.$$

It follows that

$$I(b) = \frac{\pi}{2}\sinh^{-1}(b) + C.$$

Clearly $I(0) = 0$, so $C = 0$. □

11

$$\int \frac{1}{x \sin^{-1}(\ln(x)) \sqrt{1-(\ln(x))^2}} \, \mathrm{d}x$$

Answer: $\ln|\sin^{-1}(\ln(x))| + C$.

Solution: It is easier to see if we rewrite the integral as

$$\int \frac{\frac{1}{x} \cdot \frac{1}{\sqrt{1-(\ln(x))^2}}}{\sin^{-1}(\ln(x))} \, \mathrm{d}x \, .$$

Now this is a simple chain rule, and the answer is $\ln|\sin^{-1}(\ln(x))| + C$. □

12

$$\lim_{n\to\infty}\int_0^1 |x-|x^2-|x^3-|\cdots-|x^{n-1}-|x^n||\ldots||||\,\mathrm{d}x$$

Answer: $1-\ln(2)$.

Solution: Since the integral bounds are 0 to 1, we have $x^{n-1}\geq x^n\geq 0$. The integral then simplifies to

$$\begin{aligned}&\int_0^1 (x-(x^2-(x^3-(\cdots-(x^{n-1}-(x^n))\ldots))))\,\mathrm{d}x\\&=\int_0^1 (x-x^2+x^3-\cdots+(-1)^{n+1}x^n)\,\mathrm{d}x\\&=\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\cdots+(-1)^{n+1}\frac{1}{n+1}.\end{aligned}$$

Taking the limit as $n\to\infty$ gives the result from the well-known alternating harmonic series as $1-\ln(2)$. □

13

$$\int_{|x|+|y|+|z|\leq 1} \lceil x^2+y^2+z^2 \rceil \, \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$$

Answer: $\frac{4}{3}$.

Solution: First, let us observe that in the region of integration (excluding the origin), we have

$$0 < x^2+y^2+z^2 \leq (|x|+|y|+|z|)^2 \leq 1.$$

Thus the integral is

$$\int_{|x|+|y|+|z|\leq 1} 1 \, \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z \, .$$

The region defined by $|x|+|y|+|z| \leq 1$ is a regular octahedron with vertices on the axes at unit distance from the origin. Its edge length is $\sqrt{2}$, and therefore its volume is $\frac{\sqrt{2}}{3} \cdot (\sqrt{2})^3 = \frac{4}{3}$. □

14

$$\int_0^{2\pi} \sin(\sin(x) + 2025x)\,\mathrm{d}x$$

Answer: 0.

Solution: Perform the shift $x \mapsto x - \pi$. We get

$$\begin{aligned}
&\int_0^{2\pi} \sin(\sin(x) + 2025x)\,\mathrm{d}x \\
&\quad = \int_{-\pi}^{\pi} \sin(\sin(x+\pi) + 2025(x+\pi))\,\mathrm{d}x \\
&\quad = \int_{-\pi}^{\pi} \sin(-\sin(x) + 2025x + 2025\pi)\,\mathrm{d}x \\
&\quad = -\int_{-\pi}^{\pi} \sin(-\sin(x) + 2025x)\,\mathrm{d}x
\end{aligned}$$

Observe that this final integrand is odd, therefore the answer is 0.

Alternatively, King's rule works equally well, with

$$\begin{aligned}
&\int_0^{2\pi} \sin(\sin(x) + 2025x)\,\mathrm{d}x \\
&\quad = \int_0^{2\pi} \sin(\sin(2\pi - x) + 2025(2\pi - x))\,\mathrm{d}x \\
&\quad = \int_0^{2\pi} \sin(\sin(-x) - 2025x)\,\mathrm{d}x \\
&\quad = -\int_0^{2\pi} \sin(\sin(x) + 2025x)\,\mathrm{d}x\,.
\end{aligned}$$

□

15

$$\int_0^{\pi} \sqrt{1+\sin(2x)}\,\mathrm{d}x$$

Answer: $2\sqrt{2}$.

Solution: This problem is easy in principle, but be aware of the absolute value trap! Using $1 = \sin^2(x) + \cos^2(x)$ and $\sin(2x) = 2\sin(x)\cos(x)$, we get

$$\begin{aligned}
&\int_0^{\pi} \sqrt{1+\sin(2x)}\,\mathrm{d}x \\
&= \int_0^{\pi} |\sin(x)+\cos(x)|\,\mathrm{d}x \\
&= \int_0^{\frac{3\pi}{4}} (\sin(x)+\cos(x))\,\mathrm{d}x - \int_{\frac{3\pi}{4}}^{\pi} (\sin(x)+\cos(x))\,\mathrm{d}x \\
&= 2\sqrt{2}.
\end{aligned}$$

□

16

$$\int_0^\infty \frac{1}{(x^4+1)^2}\,\mathrm{d}x$$

Answer: $\frac{3\pi}{8\sqrt{2}}$.

Solution: This is a very popular integral. Observe that

$$J := \int_0^\infty \frac{dx}{x^4+1} = \int_0^\infty \frac{x^2}{x^4+1}dx = \int_0^\infty \frac{dx}{x^2+2} = \frac{\pi}{2\sqrt{2}}.$$

Then writing $1 = (1+x^4) - x^4$ and performing IBP gives us

$$\int_0^\infty \frac{1}{x^4+1} - \frac{x^4}{(x^4+1)^2}dx = J + \left[\frac{x}{4(x^4+1)}\right]_0^\infty - \frac{1}{4}J = \frac{3}{4}J = \frac{3\pi}{8\sqrt{2}}.$$

□

17

$$\int_1^\infty \frac{1}{x+nx^n}\,\mathrm{d}x, \text{ where } n>1$$

Answer: $\dfrac{1}{1-n}\ln\left(\dfrac{n}{n+1}\right)$.

Solution: The trick here is to divide both numerator and denominator by x^n, to get

$$\int_1^\infty \frac{1}{x+nx^n}\,\mathrm{d}x = \int_1^\infty \frac{x^{-n}}{x^{-n+1}+n}\,\mathrm{d}x\,.$$

We recognise the numerator as a scaled derivative of the denominator, and thus the integral evaluates to

$$\int_1^\infty \frac{x^{-n}}{x^{-n+1}+n}\,\mathrm{d}x = \left[\frac{1}{-n+1}\ln\left(x^{-n+1}+n\right)\right]_1^\infty = \frac{1}{1-n}\ln\left(\frac{n}{n+1}\right).$$

Note that $n>1$ was required to ensure convergence. □

18

$$\int_0^{\frac{\pi}{2}} \frac{x}{\tan(x)}\,\mathrm{d}x$$

Answer: $\frac{\pi}{2}\ln(2)$.

Solution: Use integration by parts to get

$$\int_0^{\frac{\pi}{2}} x\cot(x)\,\mathrm{d}x = \left[x\ln(\sin(x))\right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \ln(\sin(x))\,\mathrm{d}x\,.$$

The first term is 0 and the second is the famous log-sine integral with value $-\frac{\pi}{2}\ln(2)$, so the final result is $\frac{\pi}{2}\ln(2)$. □

19

$$\int_0^{2025} \frac{x}{\sqrt{x+\sqrt{x+\sqrt{x+\ldots}}}} - \sqrt{x-\sqrt{x-\sqrt{x-\ldots}}}\,\mathrm{d}x$$

Answer: 0.

Solution: Consider the infinite radicals separately. From

$$y = \sqrt{x-\sqrt{x-\sqrt{x-\ldots}}} = \sqrt{x-y}$$

we get $y = \dfrac{-1+\sqrt{1+4x}}{2}$. Similarly, from

$$z = \sqrt{x+\sqrt{x+\sqrt{x+\ldots}}} = \sqrt{x+z}$$

we get $z = \dfrac{1+\sqrt{1+4x}}{2}$. Then the integral is

$$\begin{aligned}
&\int_0^{2025} \frac{2x}{1+\sqrt{1+4x}} - \frac{-1+\sqrt{1+4x}}{2}\,\mathrm{d}x \\
&= \int_0^{2025} \frac{2x}{1+\sqrt{1+4x}} - \frac{-1+\sqrt{1+4x}}{2}\,\mathrm{d}x \\
&= \int_0^{2025} \frac{\sqrt{1+4x}-1}{2} - \frac{-1+\sqrt{1+4x}}{2}\,\mathrm{d}x \\
&= 0.
\end{aligned}$$

□

20

$$\int_0^{\frac{\pi}{2}} \int_0^1 \exp\left(t + t^{\tan(\theta)}\right) \mathrm{d}t \, \mathrm{d}\theta$$

Answer: $\frac{\pi}{4}\left(e^2 - 1\right)$.

Solution: First, border flip θ and write $t = u^{\tan(\theta)}$ to get

$$\begin{aligned} I &:= \int_0^{\frac{\pi}{2}} \int_0^1 \exp\left(t + t^{\tan(\theta)}\right) \mathrm{d}t \, \mathrm{d}\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^1 \exp\left(t + t^{\cot(\theta)}\right) \mathrm{d}t \, \mathrm{d}\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^1 \exp\left(u^{\tan(\theta)} + u\right) \tan(\theta) u^{\tan(\theta)-1} \, \mathrm{d}u \, \mathrm{d}\theta \,. \end{aligned}$$

Adding the two expressions, we have

$$I = \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_0^1 \exp\left(t + t^{\tan(\theta)}\right) \left(1 + \tan(\theta) t^{\tan(\theta)-1}\right) \mathrm{d}t \, \mathrm{d}\theta \,.$$

Finally, writing $u = t + t^{\tan(\theta)}$, we get

$$I = \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_0^2 \exp(u) \, \mathrm{d}u \, \mathrm{d}\theta = \frac{\pi}{4}\left(e^2 - 1\right) .$$

□

21

$$\int_0^1 \dots \int_0^1 \max(x_1, \dots, x_{2025}) \, \mathrm{d}x_1 \dots \mathrm{d}x_{2025}$$

Answer: $\dfrac{2025}{2026}$.

Solution: For each $1 \leq i \leq 2025$, let V_i denote the region of $[0,1]^{2025}$ in which $x_i = \max(x_1, \dots, x_{2025})$. Then the integral is

$$\begin{aligned}
&\int_0^1 \dots \int_0^1 \max(x_1, \dots, x_{2025}) \, \mathrm{d}x_1 \dots \mathrm{d}x_{2025} \\
&= \sum_{i=1}^{2025} \int_{V_i} x_i \, \mathrm{d}x_1 \dots \mathrm{d}x_{2025} \\
&= \sum_{i=1}^{2025} \int_0^1 x_i \, \mathrm{Vol}([0, x_i]^{2024}) \, \mathrm{d}x_i \\
&= \sum_{i=1}^{2025} \int_0^1 x_i^{2025} \, \mathrm{d}x_i \\
&= \frac{2025}{2026}.
\end{aligned}$$

Here is an alternate solution. This year's edition of the founder's favourite problem will be familiar to probabilists who have done order statistics. Let $X_1, \dots, X_{2025}$ be i.i.d. random variables with uniform distribution on $[0,1]$. Then the integral is the well-known result

$$\mathbb{E}[\max(X_1, \dots, X_{2025})] = \frac{2025}{2026}.$$

□

22

$$\int_0^\infty \left(\frac{x}{(2^x-1)^2} - \frac{4x}{(4^x-1)^2} \right) \mathrm{d}x$$

Answer: $\dfrac{1}{\ln(2)}$.

Solution: This is a very interesting problem. Upon multiplying by $1 = \dfrac{x}{x}$, the integral becomes

$$\int_0^\infty \frac{1}{x} \left(\frac{x^2}{(2^x-1)^2} - \frac{(2x)^2}{(2^{2x}-1)^2} \right) \mathrm{d}x\,,$$

which is of Frullani form with $f(x) := \dfrac{x^2}{(2^x-1)^2}$ and limits

$$\begin{aligned} f(0^+) &= \lim_{x\to 0} \frac{x^2}{(x\ln(2)+\cdots)^2} = \frac{1}{(\ln(2))^2}, \\ f(\infty) &= 0. \end{aligned}$$

Then the value of the integral is $\dfrac{1}{(\ln(2))^2}\ln(2) = \dfrac{1}{\ln(2)}$.

What makes this interesting is the following naïve calculation: "Observe" that

$$\int_0^\infty \frac{4x}{(4^x-1)^2}\,\mathrm{d}x = \int_0^\infty \frac{y}{(2^y-1)^2}\,\mathrm{d}y\,.$$

by substitution $y = 2x$. Then splitting the original integral and transforming as above, we get 0. This second calculation is false. Splitting the integral is invalid, as the two resulting integrals now diverge. □

23

$$\int_1^e \left(\sqrt{\ln(x)} + \frac{1}{e-1} \exp\left(\left(\frac{x-1}{e-1} \right)^2 \right) \right) \mathrm{d}x$$

Answer: e.

Solution: Although this problem looks incredibly scary, it is in fact just an inverse function problem. Split the integral as

$$\int_1^e \sqrt{\ln(x)}\,\mathrm{d}x + \int_1^e \frac{1}{e-1} \exp\left(\left(\frac{x-1}{e-1} \right)^2 \right) \mathrm{d}x\,.$$

The second integral simplifies by

$$\int_1^e \frac{1}{e-1} \exp\left(\left(\frac{x-1}{e-1} \right)^2 \right) \mathrm{d}x = \int_0^{e-1} \frac{1}{e-1} \exp\left(\left(\frac{x}{e-1} \right)^2 \right) \mathrm{d}x = \int_0^1 e^{y^2}\,\mathrm{d}y\,.$$

The functions $f(x) = \sqrt{\ln(x)}$ and $g(x) = \exp(x^2)$ are inverses of each other, and thus the answer is $ef(e) - f(1) = e$. □

24

$$\int_0^\infty \frac{x^2 + \frac{2^{-\frac{x}{6}}}{7} - \frac{2^{-\frac{x}{7}}}{6}}{(2^{-\frac{x}{7}} + 36x^2)(2^{-\frac{x}{6}} + 49x^2)}\,\mathrm{d}x$$

Answer: 0.

Solution: Observe that

$$\begin{aligned}
&\int_0^\infty \frac{x^2 + \frac{2^{-\frac{x}{6}}}{7} - \frac{2^{-\frac{x}{7}}}{6}}{(2^{-\frac{x}{7}} + 36x^2)(2^{-\frac{x}{6}} + 49x^2)}\,\mathrm{d}x \\
&= \int_0^\infty \frac{\frac{1}{7}(2^{-\frac{x}{6}} + 49x^2) - \frac{1}{6}(2^{-\frac{x}{7}} + 36x^2)}{(2^{-\frac{x}{7}} + 36x^2)(2^{-\frac{x}{6}} + 49x^2)}\,\mathrm{d}x \\
&= \int_0^\infty \left(\frac{\frac{1}{7}}{2^{-\frac{x}{7}} + 36x^2} - \frac{\frac{1}{6}}{2^{-\frac{x}{6}} + 49x^2} \right) \mathrm{d}x \\
&= \int_0^\infty \frac{1}{x} \left(\frac{\frac{6x}{42}}{2^{-\frac{6x}{42}} + (6x)^2} - \frac{\frac{7x}{42}}{2^{-\frac{7x}{42}} + (7x)^2} \right) \mathrm{d}x\,.
\end{aligned}$$

This is of Frullani type with $f(x) = \dfrac{\frac{x}{42}}{2^{-\frac{x}{42}} + x^2}$. It is clear that $f(0) = f(\infty) = 0$. Hence the integral evaluates to 0. □

25

$$\int_0^{25} \sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)\sqrt{\dots}}}}\,\mathrm{d}x$$

Answer: $\dfrac{675}{2}$.

Solution: The trick here is to recognise the integrand as the limit

$$\begin{aligned}&\sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)\sqrt{\dots}}}}\\ &\quad=\lim_{n\to\infty}\sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)\sqrt{\cdots+(x+n-1)\sqrt{1+(x+n)(x+n+2)}}}}}.\end{aligned}$$

Observe that since $x>0$,

$$\begin{aligned}&\sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)\sqrt{\cdots+(x+n-1)\sqrt{1+(x+n)(x+n+2)}}}}}\\ &=\sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)\sqrt{\cdots+(x+n-2)\sqrt{1+(x+n-1)(x+n+1)}}}}}\\ &=\dots\\ &=\sqrt{1+x(x+2)}\\ &=x+1.\end{aligned}$$

Hence the nested radical evaluates to $x+1$. The answer is therefore $\displaystyle\int_0^{25}(x+1)\,\mathrm{d}x=\frac{675}{2}$. □

26

$$\int_0^1 \frac{x(x-1)}{(x+1)\ln(x)}\,\mathrm{d}x$$

Answer: $\ln\left(\frac{4}{\pi}\right)$.

Solution: This is an exercise in how well do you know your tricks:

$$\begin{aligned}
&\int_0^1 \frac{x(x-1)}{(x+1)\ln(x)}\,\mathrm{d}x \\
&= \int_0^1 \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}(x-1)}{\ln(x)}\,\mathrm{d}x \\
&= \sum_{n=0}^{\infty} (-1)^n \int_0^{\infty} \frac{e^{-(n+2)t}(1-e^{-t})}{t}\,\mathrm{d}t && \text{using } x = e^{-t} \\
&= \sum_{n=0}^{\infty} (-1)^n \ln\left(\frac{n+3}{n+2}\right) && \text{using Frullani} \\
&= \sum_{\substack{n=3 \\ n \text{ odd}}}^{\infty} \ln\left(\frac{n^2}{n^2-1}\right) \\
&= \ln\left(\frac{4}{\pi}\right) && \text{by Wallis product.}
\end{aligned}$$

□

27

$$\int \frac{2^x}{(1+\sqrt{3})^x + (2+\sqrt{3})^x}\,\mathrm{d}x$$

Answer: $\dfrac{(\sqrt{3}-1)^x - \ln\left(1+(\sqrt{3}-1)^x\right)}{\ln\left(\sqrt{3}-1\right)} + C.$

Solution: The trick here is to recognise a square of a surd.

$$\begin{aligned}
&\int \frac{2^x}{(1+\sqrt{3})^x + (2+\sqrt{3})^x}\,\mathrm{d}x \\
&= \int \frac{1}{\left(\frac{1+\sqrt{3}}{2}\right)^x + \left(\frac{2+\sqrt{3}}{2}\right)^x}\,\mathrm{d}x \\
&= \int \frac{1}{\left(\frac{1+\sqrt{3}}{2}\right)^x + \left(\frac{1+\sqrt{3}}{2}\right)^{2x}}\,\mathrm{d}x \\
&= \frac{1}{\ln\left(\frac{1+\sqrt{3}}{2}\right)} \int \frac{1}{u^2(u+1)}\,\mathrm{d}u \qquad \text{using } u = \left(\frac{1+\sqrt{3}}{2}\right)^x \\
&= \frac{1}{\ln\left(\frac{1+\sqrt{3}}{2}\right)} \int \frac{1}{u+1} - \frac{1}{u} + \frac{1}{u^2}\,\mathrm{d}u \\
&= \frac{1}{\ln\left(\frac{1+\sqrt{3}}{2}\right)} \left(\ln\left(1+\frac{1}{u}\right) - \frac{1}{u}\right) + C \\
&= \frac{(\sqrt{3}-1)^x - \ln\left(1+(\sqrt{3}-1)^x\right)}{\ln\left(\sqrt{3}-1\right)} + C.
\end{aligned}$$

□

28

$$\int_{-90}^{90} \operatorname{mrt}(x^3 - (3+a)x^2 + (4+2a)x - (2+a))\,\mathrm{d}a,$$

where $\operatorname{mrt}(f(x))$ is 0 if $f(x)$ has less than two distinct real roots, and otherwise is the smallest distance between distinct real roots of $f(x)$.

Answer: $90(45 - \sqrt{2024}) - 2\ln\left(45 - \sqrt{2024}\right) - 3\ln(2)$.

Solution: Observe that $x = 1$ is a root of the cubic, yielding

$$x^3 - (3+a)x^2 + (4+2a)x - (2+a) = (x-1)(x^2 - (2+a)x + (2+a)).$$

Hence, the other two roots are

$$x = 1 + \frac{a \pm \sqrt{a^2-4}}{2}.$$

Hence, we require $|a| \geq 2$ for mrt to be non-zero. We do not need to worry about $a = \pm 2$ since these are singletons. Otherwise, we have three distinct roots. Observe that for $a > 2$,

$$1 < 1 + \frac{a - \sqrt{a^2-4}}{2} < 1 + \frac{a + \sqrt{a^2-4}}{2},$$

and for $a < 2$

$$1 > 1 + \frac{a + \sqrt{a^2-4}}{2} > 1 + \frac{a - \sqrt{a^2-4}}{2}.$$

Hence, for $a > 2$

$$\begin{aligned}\left(1 + \frac{a + \sqrt{a^2-4}}{2}\right) - \left(1 + \frac{a - \sqrt{a^2-4}}{2}\right) &> \left(1 + \frac{a - \sqrt{a^2-4}}{2}\right) - 1 \\ \iff \sqrt{a^2-4} &> \frac{a - \sqrt{a^2-4}}{2} \\ \iff a &> \frac{3}{\sqrt{2}},\end{aligned}$$

and for $a < 2$

$$\begin{aligned}\left(1 + \frac{a + \sqrt{a^2-4}}{2}\right) - \left(1 + \frac{a - \sqrt{a^2-4}}{2}\right) &> 1 - \left(1 + \frac{a + \sqrt{a^2-4}}{2}\right) \\ \iff \sqrt{a^2-4} &> -\frac{a + \sqrt{a^2-4}}{2} \\ \iff a &< -\frac{3}{\sqrt{2}}.\end{aligned}$$

Hence, the integral is

$$\begin{aligned}&\int_{-90}^{-\frac{3}{\sqrt{2}}} \frac{-a - \sqrt{a^2-4}}{2}\,\mathrm{d}a + \left(\int_{-\frac{3}{\sqrt{2}}}^{-2} + \int_{2}^{\frac{3}{\sqrt{2}}}\right) \sqrt{a^2-4}\,\mathrm{d}a + \int_{\frac{3}{\sqrt{2}}}^{90} \frac{a - \sqrt{a^2-4}}{2}\,\mathrm{d}a \\ &= \left(2\int_{2}^{\frac{3}{\sqrt{2}}} - \int_{\frac{3}{\sqrt{2}}}^{90}\right) \sqrt{a^2-4}\,\mathrm{d}a + \int_{\frac{3}{\sqrt{2}}}^{90} a\,\mathrm{d}a\,.\end{aligned}$$

Letting

$$F(a) := \int \sqrt{a^2-4}\,\mathrm{d}a = \frac{a\sqrt{a^2-4}}{2} - 2\ln\left(a+\sqrt{a^2-4}\right),$$

(deduced by your favourite method), we see the answer is

$$\begin{aligned}
&-F(90) + 3F\left(\frac{3}{\sqrt{2}}\right) - 2F(2) + \frac{16191}{4} \\
&= -\left(90\sqrt{2024} - 2\ln\left(90+2\sqrt{2024}\right)\right) + 3\left(\frac{3}{4} - 3\ln(2)\right) - 2\left(-2\ln(2)\right) + \frac{16191}{4} \\
&= 90(45-\sqrt{2024}) - 2\ln\left(45-\sqrt{2024}\right) - 3\ln(2).
\end{aligned}$$

□

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$$\int_0^{124} (f(x))^2 \, \mathrm{d}x, \text{ where } f : [0,\infty) \to \mathbb{R} \text{ satisfies } (f(x))^3 + xf(x) = 125 \text{ for all } x \in [0,\infty).$$

Answer: 812.

Solution: The idea is to transform this into a $\mathrm{d}f$ problem. Observe that f is differentiable (solvable by Cardano's formula and can be shown that only one such f exists), so differentiating the functional equation gives

$$\frac{\mathrm{d}f}{\mathrm{d}x} \cdot 3f^2 + f + \frac{\mathrm{d}f}{\mathrm{d}x} \cdot x = 0.$$

Further, f can never be 0 (it would contradict the functional equation), therefore

$$-\left(3f + \frac{x}{f}\right) \mathrm{d}f = \mathrm{d}x\,.$$

By inspection, we have $f(0) = 5$ and $f(124) = 1$. The integral is then

$$\int_0^{124} f^2 \, \mathrm{d}x = \int_1^5 f^2 \left(3f + \frac{x}{f}\right) \mathrm{d}f = \int_1^5 \left(3f^3 + xf\right) \mathrm{d}f = \int_1^5 \left(2f^3 + 125\right) \mathrm{d}f = 812.$$

Remark: The function $f(x)$ is explicitly given by

$$f(x) = \sqrt[3]{\frac{125}{2} + \sqrt{\left(\frac{125}{2}\right)^2 + \left(\frac{x}{3}\right)^3}} + \sqrt[3]{\frac{125}{2} - \sqrt{\left(\frac{125}{2}\right)^2 + \left(\frac{x}{3}\right)^3}}.$$

□

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$$\int_1^2 \frac{\tan^{-1}(1+x)}{x}\,\mathrm{d}x$$

Answer: $\frac{3\pi\ln(2)}{8}$.

Solution: Use the border flip substitution $x \mapsto \frac{2}{x}$ to obtain

$$I := \int_1^2 \frac{\tan^{-1}(1+x)}{x}\,\mathrm{d}x = \int_1^2 \frac{\tan^{-1}\left(1+\frac{2}{x}\right)}{x}\,\mathrm{d}x\,.$$

Then by adding and applying the inverse tangent summation formula, we get

$$\begin{aligned} I &= \frac{1}{2}\int_1^2 \frac{\tan^{-1}(1+x)+\tan^{-1}\left(1+\frac{2}{x}\right)}{x}\,\mathrm{d}x \\ &= \frac{1}{2}\int_1^2 \frac{\pi+\tan^{-1}(-1)}{x}\,\mathrm{d}x \\ &= \int_1^2 \frac{3\pi}{8x}\,\mathrm{d}x \\ &= \frac{3\pi}{8}\ln(2). \end{aligned}$$

Alternative: Perform integration by parts. We have

$$\begin{aligned} \int_1^2 \frac{\tan^{-1}(1+x)}{x}\,\mathrm{d}x &= \left[\ln(x)\tan^{-1}(1+x)\right]_1^2 - \int_1^2 \frac{\ln(x)}{(1+x)^2+1}\,\mathrm{d}x \\ &= \ln(2)\tan^{-1}(3) - \int_1^2 \frac{\ln(x)}{(1+x)^2+1}\,\mathrm{d}x \end{aligned}$$

Let us evaluate this resulting integral. Consider the more general form

$$I(a) = \int_1^2 \frac{\ln\left(\frac{x}{a}\right)}{(1+x)^2+1}\,\mathrm{d}x\,,$$

and we shall find a value of $a > 0$ such that $I(a) = 0$. Let $u = \frac{x}{a}$. Then we have

$$I(a) = \int_{\frac{1}{a}}^{\frac{2}{a}} \frac{a\ln(u)}{a^2u^2+2au+2}\,\mathrm{d}u\,.$$

Observe that performing $u \mapsto \frac{1}{u}$ gives

$$I(a) = \int_{\frac{a}{2}}^{a} \frac{-a\ln(u)}{\left(1+\frac{a}{u}\right)^2+1}\cdot\frac{1}{u^2}\,\mathrm{d}u = \int_{\frac{a}{2}}^{a} \frac{-a\ln(u)}{2u^2+2au+a^2}\,\mathrm{d}u\,.$$

Now observe that $I(\sqrt{2}) = -I(\sqrt{2})$, so $I(\sqrt{2}) = 0$. Finally, we have

$$\begin{aligned} \int_1^2 \frac{\ln(x)}{(1+x)^2+1}\,\mathrm{d}x &= \ln\left(\sqrt{2}\right)\int_1^2 \frac{1}{(1+x)^2+1}\,\mathrm{d}x + \int_1^2 \frac{\ln\left(\frac{x}{\sqrt{2}}\right)}{(1+x)^2+1}\,\mathrm{d}x \\ &= \ln\left(\sqrt{2}\right)\left[\tan^{-1}(1+x)\right]_1^2 + 0 \\ &= \frac{\ln(2)(\tan^{-1}(3)-\tan^{-1}(2))}{2}. \end{aligned}$$

Putting the results together, the required value by inverse tangent addition formula is

$$\frac{\ln(2)(\tan^{-1}(3)+\tan^{-1}(2))}{2} = \frac{\ln(2)(\tan^{-1}(-1)+\pi)}{2} = \frac{3\pi\ln(2)}{8}.$$

□